AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2014

Computer Science Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

Write an algorithm to calculate and print the sum, product and average of 4 consecutive odd numbers.

Q.2. (Total 5 Marks)

A shop manager has four boys delivering newspapers to their customers. Each boy delivers three different newspapers (RECORDER, MAIL and ARGUS). The shop manager uses a spreadsheet to record the deliveries as shown below:

	A	B	C	D	E
1	**NAME**	**RECORDER**	**MAIL**	**ARGUS**	**PAID**
2	FARHAN	21	30	19	True
3	MUNIR	24	24	22	False
4	ZAHID	19	16	26	True
5	AHMED	18	12	23	False
6	TOTAL				

a. Name the data type in cells A2 and D3. (2 Marks)

_____

_____

b. The shop manager wants to add the data in cells B6, C6 and D6. What data type would you suggest for this? (1 Mark)

_____

c. Categorize the following operators into relational or logical. (2 Marks)

AND **< >** **≥** **OR** **<=** **NOT**

Relational Operator	Logical Operator

Q.3. (Total 5 Marks)

Consider the following GW BASIC program.

```
10 A = 1
20 WHILE A < = 3
30 FOR B = 1 TO 3
40 IF A = B THEN PRINT A,B
50 A = A+1
60 WEND A
70 NEXT B
```

a. Remove errors from the given program. (2 Marks)

b. Write the output of the given program. (3 Marks)

Q.4. (Total 4 Marks)

Define the term array. How are one and two dimensional arrays helpful?

Q.5. (Total 3 Marks)

Draw a square in high resolution mode using DRAW statement.

Q.6. (Total 3 Marks)

Simplify the following Boolean expression using the given Karnaugh map.

F = X.$\overline{Y}$.$\overline{Z}$ +X.$\overline{Y}$.Z +X.Y.Z + X.Y.$\overline{Z}$

	$\overline{Y}$	$\overline{Y}$	Y	Y
$\overline{X}$				
X				
	$\overline{Z}$	Z	Z	$\overline{Z}$

Q.7. (Total 8 Marks)

EITHER

a. The word ‘robot’ comes from the Czech word 'robota' which means 'forced work or labour'. Today, we use the word ‘robot’ to mean any man-made machine that can perform work or other actions normally performed by humans, either automatically or by remote control.

i. Write THREE applications of robots in the car industry.
ii. Write THREE general advantages of using robots.
iii. Write TWO general limitations of using robots.

OR

b. Security is particularly challenging when computers are connected to the internet.

What does the term ‘security’ mean in the web world? Describe at least SIX ways to secure personal data from unauthorized users.

Q.8. (Total 7 Marks)

EITHER

a. Write a GW BASIC program to read 10 values in a numeric array using INPUT statement then find and print the largest value from them.
(Note: Use nested loops and appropriate conditional statements)

OR

b. Write a GW BASIC program to print the following information using READ DATA statement.

X	Mean Deviation = X – $\bar{X}$
5	5 - 8 = -3
10	10 – 8 = 2
9	9 – 8 = 1
8	8 – 8 =0
	Sum of Mean Deviation = 0

Where $\overline{X} = \frac{5+10+9+8}{4}$

The output may look like

X	X – Mean
5	-3
10	2
9	1
8	0

Sum of Mean Deviation = 0

END OF PAPER

Please use this page for rough work

Please use this page for rough work